Sehati — Functional Requirements Document

Private Clinics Version

1. System Overview

System Name: Sehati System Type: Healthcare Management & Unified Medical Records Platform Target Users: Private Clinics & Doctors

Main Purpose:

Sehati is a digital platform that enables private clinics and doctors to manage patients' medical records digitally and allows each patient to have a unified medical profile that can be accessed securely by authorized healthcare providers.

Main Value Proposition:

One Patient, One Medical History, Better Healthcare.

2. System Users

2.1 Patient

The patient can:

- Create an account.
- Log in to the system.
- View personal medical information.
- View medical history.
- View prescriptions.
- View tests and scans.
- Track appointments.
- Track follow-ups.
- Manage the Emergency Medical Profile.
- Control medical record sharing permissions.

2.2 Doctor

The doctor can:

- Create a doctor account.
- Manage the professional profile.
- Search for patients.
- Access authorized patient medical records.
- Add diagnoses.
- Add treatment plans.

- Create prescriptions.
- Add clinical notes.
- Schedule follow-ups.
- Review previous medical history.

2.3 Clinic Receptionist

- Register patients.
- Manage appointments.
- Search for patients.
- Create visits and check-in records.
- Manage the patient queue.
- Confirm appointments.
- Cancel or reschedule appointments.

Important: The receptionist must not have unrestricted access to sensitive medical information. Permissions should be limited to the administrative information required for their duties.

2.4 Clinic Admin / Clinic Owner

The clinic administrator can:

- Manage clinic information.
- Add and manage doctors.
- Add and manage receptionists.
- Manage staff permissions.
- Monitor appointments.
- Monitor clinic activity.
- View reports and dashboards.

2.5 System Admin

The system administrator is responsible for:

- Managing users.
- Managing clinics.
- Verifying accounts.
- Managing security settings.
- Monitoring the system.
- Reviewing audit logs.

3. Main System Modules

The system will consist of the following modules:

1. Authentication & Registration
2. Patient Management
3. Doctor Management
4. Clinic Management

5. Unified Patient ID
6. Medical Records
7. Visits Management
8. Diagnosis & Treatment
9. Prescriptions
10. Tests & Scans
11. Appointments
12. Follow-ups
13. Emergency Profile
14. Medical Record Sharing
15. Notifications
16. Reports & Dashboard
17. Permissions Management
18. Audit Logs

4. Authentication & Registration

FR-01 — Patient Registration

Actor: Patient

The patient can create an account using the following information:

- Full Name
- Phone Number
- Email
- Date of Birth
- Gender
- Password

Registration Flow:

Create Account → Verify Phone/Email → Generate Unified Patient ID → Create Medical Profile

Expected Result:

A patient profile is created and linked to a unique Unified Patient ID.

5. Unified Patient ID

FR-02 — Generate Unified Patient ID

Every patient receives a unique identifier.

Example: `SH-2026-001245`

The patient can use this ID when visiting any clinic registered on Sehati.

Business Rules:

- A patient must not have more than one Unified Patient ID.
- Each ID must be unique.
- The ID must remain unchanged after creation.
- The system should prevent duplicate patient profiles whenever possible.
- Potential duplicate profiles must be reviewed before any records are merged.

6. Clinic Registration

FR-03 — Clinic Registration

Actor: Clinic Admin

The clinic administrator registers the clinic using the following information:

- Clinic Name
- Clinic Type / Specialty
- Address
- Phone
- Email
- Working Hours
- License / Verification Information

Verification Flow:

Pending Verification → Admin Review → Approved / Rejected

Expected Result:

The clinic becomes eligible to use the system's clinic features after successful verification and approval.

7. Doctor Management

FR-04 — Add Doctor

Actor: Clinic Admin

The clinic administrator can add and manage doctors.

Doctor Information:

- Name
- Specialty
- Phone
- Email
- Medical License Information
- Working Hours
- Doctor Status

Doctor Status Options:

- Pending
- Active
- Suspended

A doctor may be associated with more than one clinic. Therefore, the relationship between doctors and clinics should be managed separately.

8. Patient Management

FR-05 — Add Patient to Clinic

When a patient visits a clinic, the system supports two options.

Option 1: The patient provides their Unified Patient ID.

Option 2: The receptionist searches for the patient using:

- Phone Number
- Unified Patient ID

If the patient already exists:

Search Patient → Verify Identity → Check Access Permissions → Open Authorized Patient Profile

If the patient does not exist:

Create Patient Profile → Generate Unified Patient ID

Business Rules:

- The system should avoid creating duplicate patient profiles.
- Patient identity must be verified before accessing medical records.
- Access to medical information must follow the patient's permissions and the system's access-control rules.

9. Medical Record

FR-06 — Patient Medical Profile

The system maintains a unified medical profile for each patient.

A. Personal Information

- Full Name
- Age
- Gender
- Contact Information

- Address

B. Medical Information

- Medical History
- Allergies
- Previous Diagnoses
- Treatments
- Current and Previous Medications
- Medical Documents
- Lab Results
- Scans
- Prescriptions
- Medical Reports

Expected Result:

All relevant medical information is linked to the patient's unified profile and can be accessed only by authorized users.

10. Patient Visit

FR-07 — Create New Visit

Actors: Doctor / Receptionist

Each patient visit must have a separate Visit Record.

Visit Data:

- Patient
- Doctor
- Clinic
- Date & Time
- Chief Complaint
- Diagnosis
- Treatment
- Prescription
- Doctor Notes
- Follow-up Date
- Visit Status

Visit Workflow:

Patient Check-in-Create Visit - Doctor Consultation - Diagnosis - Treatment – Prescription-Follow-up - Save Medical Record

Expected Result:

The visit is saved and linked to the patient's medical history, allowing authorized doctors to review it later.

11. Diagnosis

FR-08 — Add Diagnosis

Actor: Doctor

The doctor can add a diagnosis to the patient's visit.

Diagnosis Information:

- Diagnosis Name
- Symptoms
- Clinical Notes
- Diagnosis Date

Expected Result:

The diagnosis is saved in the medical history and linked to the relevant visit.

12. Treatment

FR-09 — Add Treatment

Actor: Doctor

The doctor can record a treatment plan for the patient.

Treatment Information:

- Treatment Plan
- Medication, if applicable
- Instructions
- Duration
- Start Date
- End Date
- Notes

Expected Result:

The treatment plan is saved and linked to the relevant visit.

13. Digital Prescription

FR-10 — Create Prescription

Actor: Doctor

The doctor can create a digital prescription for the patient.

Prescription Fields:

- Medication Name
- Dosage
- Frequency
- Duration
- Instructions
- Notes

A prescription may contain multiple medications, so the database should separate the prescription itself from its individual medication items.

Expected Result:

Prescription → Patient Medical Record

The patient can view the prescription through their account.

14. Tests & Scans

FR-11 — Add Test / Scan

Actor: Doctor

The doctor can record a test or scan request and add the results when they become available.

Examples:

- Blood Test
- X-Ray
- MRI
- CT Scan
- Ultrasound

Test / Scan Information:

- Test or Scan Name / Type
- Date
- Result
- Notes
- Uploaded Report or File

Expected Result:

The test or scan information and its available results are saved in the patient's medical record.

15. Appointments

FR-12 — Appointment Management

The appointment module provides different functions according to the user's role.

Patient can:

- Request an appointment.
- View appointments.
- Cancel appointments.
- Reschedule appointments.

Receptionist can:

- Create appointments.
- Confirm appointments.
- Cancel appointments.
- Reschedule appointments.
- Manage the patient queue.

Doctor can:

- View the daily schedule.
- View upcoming appointments.

Appointment Status:

- Pending
- Confirmed
- Completed
- Cancelled
- NoShow

Expected Result:

The system maintains appointment information and updates the status whenever an appointment is created, confirmed, rescheduled, completed, or cancelled.

16. Follow-up

FR-13 — Follow-up Management

Actor: Doctor

The doctor can schedule a follow-up for the patient.

Follow-up Information:

- Follow-up Date
- Follow-up Reason
- Notes

- Status

Expected Result:

The follow-up is linked to the patient, doctor, and relevant visit. The system sends a reminder to the patient according to the configured notification settings.

17. Emergency Medical Profile

FR-14 — Emergency Profile

The patient has an Emergency Medical Profile containing essential information that may be useful in emergency situations.

Emergency Profile Information:

- Allergies
- Current Medications
- Important Medical Conditions
- Blood Type, if known
- Emergency Contact Name
- Emergency Contact Phone

Expected Result:

The patient can manage and update the emergency profile.

Access to this information must follow clearly defined security rules. Any emergency access mechanism should be limited, logged, and designed according to applicable privacy and safety requirements.

18. Medical Record Sharing

FR-15 — Grant Access

Medical record sharing is one of the most important features of Sehati.

The patient can grant a doctor or clinic permission to access their medical record.

Access Flow:

Patient Searches for Clinic / Doctor ↓ Access Request ↓ Patient Approval ↓ Authorized Doctor Can Access the Permitted Record

Access Types:

1. View Only: The doctor can view the permitted medical information.
2. View + Add Medical Records: The doctor can view the permitted information and add new medical records within the authorized scope.

Business Rules:

- Access must not be granted before the required approval is recorded.
- The patient should be able to review active permissions.
- Access permissions may have an expiration date.
- The patient should be able to revoke access, subject to applicable legal and medical record-retention requirements.
- All access requests, approvals, rejections, revocations, and sensitive record activities should be logged.

19. Permissions

The system must implement Role-Based Access Control (RBAC) to ensure that each user can perform only the actions permitted for their role.

Action	Patient	Doctor	Receptionist	Clinic Admin
View Own Profile	Yes	No	No	Limited
View Medical History	Own Record	With Permission	No	Limited
Add Diagnosis	No	Yes	No	No
Add Prescription	No	Yes	No	No
Manage Appointments	Own Appointments	Own Schedule	Clinic Schedule	Clinic-wide
Manage Doctors	No	No	No	Yes
Manage Receptionists	No	No	No	Yes
Manage Sharing Permissions	Own Record	No	No	No

Note: The System Admin has separate administrative permissions for platform operations. Administrative access should not automatically grant unrestricted access to patients' clinical information.

20. Notifications

The system sends notifications for the following events:

- Appointment Confirmation
- Appointment Reminder
- Appointment Cancellation
- Appointment Rescheduling
- Follow-up Reminder
- New Prescription
- Medical Record Updated
- Access Request
- Access Approved
- Access Rejected

The system should also track notification status, such as whether a notification has been read.

21. Clinic Dashboard

The Clinic Dashboard provides an overview of clinic operations.

Dashboard Information:

- Today's Appointments
- Upcoming Appointments
- Today's Patients
- Completed Visits
- Cancelled Appointments
- No-show Appointments
- Doctors
- Receptionists

22. Reports

The Clinic Admin can view reports related to clinic activity.

Reports Include:

- Number of Patients
- Number of Visits
- Number of Appointments
- Cancelled Appointments
- Follow-ups
- Doctor Activity

Database Design

23. Database Entities and Attributes

The following section defines the proposed database structure for Sehati.

Notation:

- PK = Primary Key
- FK = Foreign Key
- Unique = Unique Constraint

23.1 Users

Purpose: Stores login and authentication data.

- UserId (PK)
- Email
- PasswordHash
- Phone

- RoleId (FK)
- IsActive
- CreatedAt

23.2 Roles

Purpose: Defines the types of users in the system.

- RoleId (PK)
- RoleName

Roles:

- Patient
- Doctor
- Receptionist
- ClinicAdmin
- SystemAdmin

23.3 Patients

Purpose: Stores the patient's basic information and unified identity.

- PatientId (PK)
- UserId (FK)
- UnifiedPatientId (Unique)
- FullName
- DateOfBirth
- Gender
- Address
- EmergencyContact
- CreatedAt

23.4 Clinics

Purpose: Stores clinic information.

- ClinicId (PK)
- ClinicName
- Specialty
- Address
- Phone
- Email
- WorkingHours
- Status
- CreatedAt

23.5 Doctors

Purpose: Stores doctors' professional information.

- DoctorId (PK)
- UserId (FK)
- FullName
- Specialty
- LicenseNumber
- Phone
- Email

23.6 ClinicDoctors

Purpose: Manages the relationship between doctors and clinics because a doctor may work at multiple clinics.

- ClinicDoctorId (PK)
- ClinicId (FK)
- DoctorId (FK)
- JoinedAt
- Status

23.7 Receptionists

Purpose: Stores receptionist information and clinic affiliation.

- ReceptionistId (PK)
- UserId (FK)
- ClinicId (FK)
- FullName

23.8 MedicalRecords

Purpose: Stores the main medical record associated with each patient.

- MedicalRecordId (PK)
- PatientId (FK)
- CreatedAt
- LastUpdatedAt

Business Rule: Each patient should have one main Medical Record.

23.9 Visits

Purpose: Stores each patient visit to a clinic.

- VisitId (PK)
- PatientId (FK)
- DoctorId (FK)
- ClinicId (FK)

- VisitDate
- ChiefComplaint
- Notes
- Status

23.10 Diagnoses

Purpose: Stores diagnoses associated with visits.

- DiagnosisId (PK)
- VisitId (FK)
- DiagnosisName
- Description
- DiagnosisDate

23.11 Treatments

Purpose: Stores treatment plans associated with visits.

- TreatmentId (PK)
- VisitId (FK)
- TreatmentDescription
- StartDate
- EndDate
- Instructions

23.12 Prescriptions

Purpose: Stores prescription information.

- PrescriptionId (PK)
- VisitId (FK)
- DoctorId (FK)
- PatientId (FK)
- PrescriptionDate
- Notes

23.13 PrescriptionItems

Purpose: Stores the individual medications included in each prescription.

- PrescriptionItemId (PK)
- PrescriptionId (FK)
- MedicationName
- Dosage
- Frequency
- Duration
- Instructions

23.14 Allergies

Purpose: Stores the patient's allergy information.

- AllergyId (PK)
- PatientId (FK)
- AllergyName
- Reaction
- Notes

23.15 Medications

Purpose: Stores the patient's current and previous medication information.

- MedicationId (PK)
- PatientId (FK)
- MedicationName
- Dosage
- Frequency
- StartDate
- EndDate
- Notes

23.16 Tests

Purpose: Stores laboratory and other test information.

- TestId (PK)
- PatientId (FK)
- VisitId (FK)
- TestName
- TestDate
- Result
- Notes
- FileUrl

23.17 Scans

Purpose: Stores medical imaging information.

- ScanId (PK)
- PatientId (FK)
- VisitId (FK)
- ScanType
- ScanDate
- Result
- FileUrl
- Notes

23.18 Appointments

Purpose: Stores appointment information and scheduling details.

- AppointmentId (PK)
- PatientId (FK)
- DoctorId (FK)
- ClinicId (FK)
- AppointmentDate
- StartTime
- EndTime
- Status
- Notes

Appointment Status:

- Pending
- Confirmed
- Completed
- Cancelled
- NoShow

23.19 FollowUps

Purpose: Stores follow-up appointments and plans.

- FollowUpId (PK)
- PatientId (FK)
- DoctorId (FK)
- VisitId (FK)
- FollowUpDate
- Notes
- Status

23.20 EmergencyProfiles

Purpose: Stores essential emergency medical information.

- EmergencyProfileId (PK)
- PatientId (FK)
- BloodType
- EmergencyNotes
- EmergencyContactName
- EmergencyContactPhone
- LastUpdatedAt

23.21 AccessPermissions

Purpose: Defines which doctors or clinics can access a patient's medical record.

- AccessPermissionId (PK)
- PatientId (FK)
- DoctorId (FK)
- ClinicId (FK)
- AccessType
- GrantedAt
- ExpiresAt
- Status

Access Types:

- ViewOnly
- ViewAndAddRecords

Business Rule: The permission must identify at least one access target: a doctor, a clinic, or both.

23.22 Notifications

Purpose: Stores notifications sent to users.

- NotificationId (PK)
- UserId (FK)
- Title
- Message
- Type
- IsRead
- CreatedAt

23.23 AuditLogs

Purpose: Records important and sensitive system activities.

- AuditLogId (PK)
- UserId (FK)
- PatientId (FK)
- Action
- EntityName
- EntityId
- Timestamp
- IPAddress

24. Main Database Entities

The main entities expected in the E-R Diagram are:

1. User
2. Role
3. Patient

4. Doctor
5. Clinic
6. Receptionist
7. MedicalRecord
8. Visit
9. Diagnosis
10. Treatment
11. Prescription
12. PrescriptionItem
13. Allergy
14. Medication
15. Test
16. Scan
17. Appointment
18. FollowUp
19. EmergencyProfile
20. AccessPermission
21. Notification
22. AuditLog

25. Main System Workflow

Patient Registration ↓ Generate Unified Patient ID ↓ Patient Visits Clinic ↓ Patient Provides Unified Patient ID ↓ Clinic Finds Patient ↓ Verify Identity and Access Permissions ↓ Doctor Views Authorized Medical History ↓ Create New Visit ↓ Add Diagnosis ↓ Add Treatment ↓ Create Prescription, if needed ↓ Schedule Follow-up, if needed ↓ Update Medical Record ↓ Send Patient Notification ↓ Record Relevant Activity in Audit Logs